Claude Code Usage Analytics

Key Findings from Telemetry Data

100 Engineers | 5,000 Sessions | 60 Days | 454,428 Events

Dec 3, 2025 - Feb 1, 2026

Dataset Overview

Scale, scope, and structure of the telemetry data

454,428	118,014	5,000	100	5	17
Events	API Requests	Sessions	Users	Models	Tools

Event Types Ingested

- API Requests (118,014) -- model, tokens, cost, latency per call
- Tool Decisions (151,461) -- accept/reject decisions for 17 tool types
- Tool Results (148,418) -- execution success/failure and duration
- User Prompts (35,173) -- prompt lengths (contents redacted for privacy)
- API Errors (1,362) -- error messages, status codes, retry attempts

Infrastructure

- 5 Claude models: Haiku 4.5, Opus 4.5, Opus 4.6, Sonnet 4.5, Sonnet 4.6
- 70% business-hours bias (9 AM - 6 PM), reflecting real work patterns
- Environments: 78% macOS/ARM64, 15% Linux, 5% Darwin x86, 2% Windows
- IDEs: VSCode (40%), PyCharm (20%), Warp Terminal (10%), others

Cost & Usage Patterns

Token consumption, model economics, and peak windows

$6,001	103.3M	8,983ms	1.2%
Total Cost	Total Tokens	Avg Latency	Error Rate

Cost by Model

Model	Cost	Requests
Opus 4.5	$2,194	23,693
Opus 4.6	$2,065	25,959
Sonnet 4.5	$1,366	19,698
Sonnet 4.6	$192	2,685
Haiku 4.5	$186	45,979

Cost by Practice

Practice	Cost	Requests
ML Engineering	$1,475	29,289
Frontend Eng.	$1,473	28,797
Data Engineering	$1,184	23,347
Backend Eng.	$1,176	22,908
Platform Eng.	$693	13,673

Key Insight

- Haiku handles 39% of requests but only 3% of cost -- ideal for high-volume, low-complexity tasks
- Opus models drive 71% of total cost; ML and Frontend practices are the top consumers
- Peak usage: Monday-Friday 9 AM - 5 PM, with sharp drop-offs on weekends

Developer Behavior

Tool usage patterns, prompt characteristics, and cost drivers

Most-Used Tools

Tool	Decisions	Share
Read	46,015	30.4%
Bash	43,214	28.5%
Edit	19,127	12.6%
Grep	11,565	7.6%
Glob	7,091	4.7%

Cost by Seniority

Level	Cost	% of Total
L5 (Mid-Senior)	$1,225	20.4%
L6 (Senior)	$1,321	22.0%
L4 (Mid)	$859	14.3%
L3 (Junior+)	$765	12.8%
L7+ (Staff+)	$1,199	20.0%

Prompt Characteristics

- Median prompt length ~128 characters; 90th percentile ~2,969 characters
- Short prompts dominate: 60% are under 200 characters (quick commands/questions)
- Long prompts (3,000+) represent complex architectural or multi-step requests

Key Insight

- Read + Bash account for 59% of all tool usage -- engineers primarily read code and run commands
- Tool success rates are high (93-99%), with Bash being the most failure-prone at 93.3%
- L5-L6 engineers (mid-senior to senior) drive 42% of total cost -- the most active cohort

Anomalies & Forecast

Outlier detection, error spikes, and cost projections

Anomaly Detection Highlights

- IQR method on daily cost identified 3 anomalous days exceeding the upper fence
- Z-score analysis on daily errors found 4 spike days (> 2 std deviations)
- Most error spikes correlate with rate-limit (429) bursts on Opus models
- Request-abort errors are the most frequent (52% of all errors)

Cohort Variance

Cohort	Avg Cost	Std Dev
ML Eng / L5	$87.20	$42.15
Frontend / L6	$92.50	$38.60
Backend / L4	$51.40	$28.30
Data Eng / L5	$65.80	$31.90
Platform / L3	$34.20	$19.70

7-Day Cost Forecast

Metric	Value
Method	OLS + day-of-week seasonality
Daily forecast range	$86 - $104
95% confidence interval	+/- $41
Weekly projected cost	$644
Trend direction	Stable (slight upward)

Recommendations

- Monitor Opus cost -- 71% of spend on 42% of
- requests; consider Haiku for suitable tasks
- Investigate rate-limit spikes -- they cluster
- on specific days, suggesting burst patterns
- Track L5-L6 cohort -- highest cost variance
- suggests uneven adoption or usage patterns

Summary

- The platform processes 454K events into actionable dashboards, API endpoints, and predictive models -- enabling data-driven decisions on Claude Code usage, cost optimizati